
Unit – 1 : Session – 3 : SLO – 2 | Multiprogramming / Time-Sharing Word Search + Kernel Structures

I. Word Search — Words to Find

1. MONITOR, MULTITASKING, PROCESSOR, VIEW, ROUND ROBIN, SHORTEST, SIMULTANEOUSLY, TIME SHARING
2. All eight words are located in the grid (horizontal, vertical, and diagonal directions).
3. Hidden message from the unused letters: "MULTIPROGRAMMING IMPROVES CPU UTILIZATION".

II. Kernel Structures — Reverse Word Search (place these words)

1. EXOKERNEL — minimalist kernel that exposes hardware to applications, giving them direct control.
2. HYBRID KERNEL — combines features of monolithic and microkernels; used in Windows and macOS.
3. LAYERED — organized in levels; each layer builds on services provided by the one below it.
4. MICROKERNEL — only essential functions like IPC and basic scheduling; everything else in user space.
5. MODULAR — components can be loaded and unloaded at runtime; used in Linux systems (LKMs).
6. MONOLITHIC — single large process running entirely in kernel mode with all OS services included.
7. SIMPLE — basic structure with minimal functionality, used in teaching or early systems (e.g. MS-DOS).
8. VIRTUAL MACHINE (HYPERVISOR) — abstracts physical hardware; runs multiple OSES on the same system.